

Regression Techniques

Lecture Notes

Indian Statistical Institute

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Chapter 1

Pre-requisites

1.1 Introduction to Linear Regression Models.

Data:

- Dependent variable or response (y),
- Independent variables or covariates or regressors ($x_1, x_2, \dots, x_p$)
- Usually, we have n observations on all variables.

$$\underset{\sim_n}{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} : \text{The vector of } n \text{ observations of the response.}$$

$$\begin{aligned} X_n &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}^{n \times p} : \text{Design matrix or matrix of } n \text{ observations of } p \text{ covariates} \\ &= ((x_{ij})) \end{aligned}$$

$\underset{\sim_i}{x}$: i th row of X_n provides the i th observation on all variables, $i = 1, \dots, n$.

$\underset{\sim_j}{x}$: j th column of X_n provides a vector of all observations on the j th variables, $j = 1, \dots, p$.

Examples

1. *Air-quality (UCI)*. Data on 5 metal oxide chemical sensors (CO, Non-metanic hydrocarbon, Benzene, NOx, NO₂) and the temperature, humidity, relative humidity. (See the data [here](#))

Q: Does air-quality affect the temperature, or humidity, or relative humidity?

2. *Boston Housing data (Kaggle)*. See the data description [here](#). Read the data description and identify the underlying regression-related question in the data.

Regression setup. How to relate the temperature with the air pollution indicators in the AQI data?

Usually all the variables are assumed to be random. At least the response variable (Y) is assumed to be random, and the n observations $y_{\sim n} = (y_1, \dots, y_n)^\top$ are realizations of n independent samples.

In regression, one assumes that

$$E(Y) = g(x_1, \dots, x_p), \quad \text{or,} \quad E(y|X_1 = x_1, \dots, X_p = x_p) = g(x_1, \dots, x_p),$$

according as the covariates are non-random, or random.

Linear Model

In linear model, we assume

$$E(Y) = g(x_1, \dots, x_p) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p,$$

where $\beta_0, \beta_1, \dots, \beta_p$ are unknown constants or parameters, to be estimated using the data.

Here linear means, linear in parameters. A function, which is not linear in $\{x_1, \dots, x_p\}$ can be a linear regression model. For example,

$$E(Y) = \beta_0 + \beta_1 \sin x_1 + \beta_2 e^{x_2}$$

is also a linear model.

Note: In simple linear model literature, $\text{Var}(Y) = \sigma^2$, which is free of $x_1, \dots, x_p$.

What can be said about joint distribution of $Y_1, \dots, Y_n$?

Random vectors. Let

$$Z_{\sim} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$$

be a random vector, i.e., $Z_1, \dots, Z_n$ are random variables and $Z_{\sim}$ is a stacked version of all of these random variables.

$$\begin{aligned}
f_{\tilde{Z}}(z) &= f_{Z_1, \dots, Z_n}(z_1, \dots, z_n) \\
&= \begin{cases} P(Z_1 = z_1, \dots, Z_n = z_n) & \text{for discrete random variables,} \\ \text{joint pdf of } \{Z_1, \dots, Z_p\} \text{ evaluated at } \{z_1, \dots, z_p\}. & \end{cases}
\end{aligned}$$

Expectation and Variance.

$$E(\tilde{Z}) = \begin{bmatrix} E(Z_1) \\ \vdots \\ E(Z_n) \end{bmatrix} = \underset{\sim Z}{\mu}$$

Let $\tilde{Z} \in \mathbb{R}^p$ and $\tilde{W} \in \mathbb{R}^q$ be two random vectors, then

$$\begin{aligned}
\text{Cov}(\tilde{Z}, \tilde{W}) &= ((\text{Cov}(Z_j, W_k)))^{p \times q}, \quad j = 1, \dots, p; \quad k = 1, \dots, q, \\
\text{Var}(\tilde{Z}) &= E \left[(\tilde{Z} - \underset{\sim Z}{\mu})(\tilde{Z} - \underset{\sim Z}{\mu})^\top \right] = ((\text{Cov}(Z_i, Z_j))) = \text{Cov}(\tilde{Z}, \tilde{Z}) = \text{Var}(\tilde{Z})
\end{aligned}$$

Note:

1. $E(\tilde{Z})$ is a non-random vectors, and $\text{Var}(\tilde{Z})$ is a non-random $p \times p$ matrix.
2. $\text{Var}(\tilde{Z})$ is a positive semi-definite matrix.

Some exercises:

Show the following:

1. $\text{Var}(\tilde{Z} - \beta) = \text{Var}(\tilde{Z})$.
2. $\text{Cov}(\underset{\sim}{a}^\top \tilde{Z}, \underset{\sim}{b}^\top \tilde{W}) = \underset{\sim}{a}^\top \text{Cov}(\tilde{Z}, \tilde{W}) \underset{\sim}{b}$.
3. $\text{Cov}(\underset{\sim}{X}_1 + \underset{\sim}{X}_2, \underset{\sim}{Z}_1 + \underset{\sim}{Z}_2) = \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_2) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_2)$.
4. $\text{Var}(\tilde{X} \pm \tilde{Z}) = \text{Var}(\tilde{X}) + \text{Var}(\tilde{Z}) \pm \text{Cov}(\tilde{X}, \tilde{Z}) \pm \text{Cov}(\tilde{Z}, \tilde{X})$.
5. $E(\tilde{Z}^\top A \tilde{Z}) = \underset{\sim}{\mu}^\top A \underset{\sim}{\mu} + \text{tr}(A \Sigma)$, where $E(\tilde{Z}) = \underset{\sim}{\mu}$, $\text{Var}(\tilde{Z}) = \Sigma$.

Back to the linear regression model:

If $Y_i \sim F(\mu_i, \sigma^2)$ some distribution with mean μ_i and variance σ^2 , then $\mu_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}$. Further, it is *assumed* that $Y_1, \dots, Y_n$ are independently distributed. Therefore,

$$E(\underset{\sim}{Y}_n) = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \underset{\sim}{X}_n \beta \quad \text{and} \quad \text{Var}(\underset{\sim}{Y}_n) = \sigma^2 I_n,$$

where $\underset{\sim}{\beta} = (\beta_0, \dots, \beta_p)^\top$ is the vector of regression parameters, and I_n is the n -dimensional identity matrix.

Alternative representation. One can rephrase the above model in the following way:

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p} + \varepsilon_i,$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$ for all $i = 1, \dots, n$.

1.2 Vector Differentiation

1.2.1 Gradient and Hessian

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$.

Example: i. $k = 2$, $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x + y$. ii. For general k , let $\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$, $g(\underline{x}) = x_1 + \dots + x_k \in \mathbb{R}$.

- **Gradient:**

$$\nabla g = \frac{\partial g}{\partial \underline{x}} = \begin{bmatrix} \partial g / \partial x_1 \\ \vdots \\ \partial g / \partial x_k \end{bmatrix}^{k \times 1}$$

For $g(\underline{x}) = x_1 + \dots + x_k$,

$$\nabla g = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1}$$

- **Second derivative / Hessian:** The matrix of second derivatives is called the Hessian matrix,

and is defined as

$$H(\underline{x}) = \left(\left(\frac{\partial^2 g}{\partial x_i \partial x_j} \right) \right) = \frac{\partial^2 g}{\partial \underline{x} \partial \underline{x}^\top} = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \cdots & \frac{\partial^2 g}{\partial x_1 \partial x_k} \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_2 \partial x_k} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 g}{\partial x_k \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_k^2} \end{bmatrix}^{k \times k}.$$

• For vector-valued $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$, $g(\underline{x}) = \begin{bmatrix} g_1(\underline{x}) \\ \vdots \\ g_k(\underline{x}) \end{bmatrix}$

$$\nabla g = \left[\left(\frac{\partial g_1}{\partial \underline{x}} \right) \cdots \left(\frac{\partial g_k}{\partial \underline{x}} \right) \right]^{k \times k} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_k}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_k} & \cdots & \frac{\partial g_k}{\partial x_k} \end{bmatrix}$$

Example: $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$, then

$$\nabla g = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$$

1.2.2 Use 1: Jacobian of Transformation

Recall the transformation technique of obtaining the distribution of a one-to-one function of X , when the distribution of X is known. Now, we will extend the transformation technique to multivariate change of variables problems.

Univariate: $X \sim f_X(x)$ (pdf), $Z = h(X)$, then

$$f_Z(z) = f_X(x) \Big|_{x \rightarrow z} |J_{x \rightarrow z}| = f_X(h^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|.$$

Multivariate: $\underline{X} \sim f_{\underline{X}}$ (joint pdf), $\underline{Z} = h(\underline{X})$ where $h : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is one-to-one function.

The pdf of $\underline{Z}$ at $\underline{z}$ is

$$f_{\underline{Z}}(\underline{z}) = f_{\underline{X}}(\underline{x}) \Big|_{\underline{x} \rightarrow \underline{z}} |J_{\underline{x} \rightarrow \underline{z}}| = f_{\underline{X}}(h^{-1}(\underline{z})) \left| \det \left(\frac{\partial \underline{x}}{\partial \underline{z}} \right) \right|.$$

Example 1. Let $X_i \stackrel{iid}{\sim} N(0, \sigma^2)$, $i = 1, 2$. Let

$$Z = \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}, \quad Z = AX, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$f_{\underline{Z}}(\underline{x}) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}(x_1^2 + x_2^2)\right\}$$

We have

$$\underline{X} = \begin{pmatrix} (z_1 + z_2)/2 \\ (z_1 - z_2)/2 \end{pmatrix} = B\underline{Z},$$

$$J_{x \rightarrow z} = \left| \det \left(\frac{\partial \underline{x}}{\partial \underline{z}} \right) \right| = \left| \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \right| = \frac{1}{2}.$$

Thus,

$$f_{\underline{Z}}(\underline{z}) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2} \left(\frac{(z_1 + z_2)^2}{4} + \frac{(z_1 - z_2)^2}{4} \right)\right\} \times \frac{1}{2}, \quad z_1 \in \mathbb{R}, z_2 \in \mathbb{R}, \sigma^2 > 0.$$

Identify the distribution of $\underline{Z}$.

1.2.3 Use 2: Finding maxima/minima of a multivariate function

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$ function, i.e., $g(\underline{z}) = g(z_1, \dots, z_k)$. We are interested in finding the maxima/minima of g w.r.t. $\underline{z}$.

Let g is maximized (minimized) w.r.t. $\underline{z}$ at $\underline{z}_{\sim_0}$, where $\underline{z}_{\sim_0}$ is on ‘interior point’ of the domain. The following steps may lead to the identification $\underline{z}_{\sim_0}$:

1. Differentiate g w.r.t. $\underline{z}$, and equate that to 0, $\frac{\partial g}{\partial \underline{z}} = 0$.
2. Find the points for which this equation is satisfied. Let $\underline{x}_{\sim_1}, \dots, \underline{x}_{\sim_m}$ be the points. These are called stationary points, or critical points.

[Note:] A critical point can be maxima, minima or a saddle point.

3. Next evaluate the Hessian $\frac{\partial^2 g}{\partial \underline{z} \partial \underline{z}^\top} \bigg|_{\underline{x}_{\sim_j}} = H(\underline{x}_{\sim_j})$, $j = 1, \dots, k$.

If $\underline{x}_0$ is a maxima (minima) then $H(\underline{x}_0)$ will be negative (positive) definite.

Example 2. $g : \mathbb{R}^k \rightarrow \mathbb{R}$,

$$g(\underline{x}) = \sum_{j=1}^k (x_j^4 - 4x_j^2).$$

i)

$$\nabla g = \begin{bmatrix} 4x_1^3 - 8x_1 \\ \vdots \\ 4x_k^3 - 8x_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Solutions to the first order equations: $x_j \in \{0, \sqrt{2}, -\sqrt{2}\}$. So total 3^k critical points:

$$\underline{x}_1^* = \underline{0}, \quad \underline{x}_2^* = \sqrt{2}\underline{1}, \quad \underline{x}_3^* = \begin{pmatrix} \sqrt{2} \\ \vdots \\ -\sqrt{2} \end{pmatrix}, \quad \dots, \quad x_m^* = -\sqrt{2}\underline{1}, \quad x_{m+1}^* = \begin{pmatrix} 0 \\ \sqrt{2} \\ \vdots \\ \sqrt{2} \end{pmatrix}, \quad \dots$$

ii)

$$H(\underline{x}) = \text{Diag}(12x_1^2 - 8, \dots, 12x_k^2 - 8)$$

Therefore, $H(\mathbf{0}) = -8I$: negative definite, which implies that $\mathbf{0}$ is a local maxima.

For all combinations with $x_j = \pm\sqrt{2} \forall j$, $H = 16I$: positive definite, and therefore, all these points are local minimas.

Note: Points with mix of 0 and $\pm\sqrt{2}$ give indefinite Hessian.

Some exercises:

1. $g(\underline{x}) = \|\underline{y} - A\underline{x}\|^2$ where A is a p.d. invertible matrix.
2. $g(\underline{x}) = \|\underline{x}\|^2 + \exp\{\underline{a}^T \underline{x}\}$

1.3 Multivariate Normal Distribution

Multivariate normal distribution plays a key role in multivariate analysis. To model a multivariate data, multivariate normal is most commonly used. The motivations behind this predominance can be stated below.

- [1] As already stated, providing a multivariate generalization to any univariate distribution may not be easy. The generalization may not be unique. For normal distribution easy extension is available.
- [2] The p -mvt. normal distribution is completely specified by the first 2 moments. The interpretable parameters, and availability of efficient estimators of the parameters, make multivariate normal more popular.
- [3] The fact that for jointly normally distributed variables, uncorrelatedness $\Leftrightarrow$ independence, makes verification of independence far less complicated than for normal distribution.

Defn (Multivariate Normal)

A p -variate RV $\underline{X}$ is said to follow a multivariate normal distribution with parameters $\underline{\mu}$ and $\Sigma \geq 0$ iff

$$\underline{a}^T \underline{X} \sim \mathcal{N}(\underline{a}^T \underline{\mu}, \underline{a}^T \Sigma \underline{a}) \text{ for all } \underline{a} \in \mathbb{R}^p$$

Here $\underline{\mu} \in \mathbb{R}^p$, and Σ is a $p \times p$ symmetric PSD matrix.

We write $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$ if $\underline{x}$ follows a multivariate normal with parameters $\underline{\mu}$ and Σ .

Note: $X_i \sim N(\mu_i, \sigma_{ii})$ for all $i = 1, \dots, p$ does not imply $\underline{X}$ is normal. Let $\underline{e}_i$ be the vector with all zero elements except the i -th one, which is one. If $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$ where $\underline{\mu} = (\mu_1, \dots, \mu_p)^T$ and $\Sigma = ((\sigma_{ij}))$, then

$$\underline{e}_i^T \underline{X} = X_i \sim N(\mu_i, \sigma_{ii}) \quad \forall i.$$

However, the following example shows a case where the marginals are normal but the joint is not.

Let (X, Y) is a bivariate random variable with joint pdf

$$f(x, y) = \frac{1}{2\pi} \exp\left\{-\frac{x^2 + y^2}{2}\right\} \left\{1 + xy \exp\left\{-\frac{x^2 + y^2}{2}\right\}\right\}$$

Properties :**[1] Characteristic function:**

Recall that the characteristic function (ch.fn.) of univariate normal with parameters η and σ^2 is $\exp(i\eta t - \frac{1}{2}t^2\sigma^2)$, at the point t . Thus,

$$\phi_{\underline{X}}(\underline{t}) = E[\exp\{i\underline{t}^T \underline{X}\}] = E[\exp\{iY\} \mid Y \sim \mathcal{N}(\underline{t}'\underline{\mu}, \underline{t}'\Sigma\underline{t})] = \exp\left(i\underline{t}'\underline{\mu} - \frac{1}{2}\underline{t}'\Sigma\underline{t}\right).$$

[2] Linear transformation:

Let $\underline{Y} = A\underline{X} + \underline{b}$, where A is a $q \times p$ matrix, and $\underline{b} \in \mathbb{R}^q$. Then $\underline{y} \sim \mathcal{N}_q(A\underline{\mu} + \underline{b}, A\Sigma A')$

Proof: The ch fn. of $\underline{Y}$ is

$$\begin{aligned} \phi_{\underline{Y}}(\underline{t}) &= E[\exp\{i\underline{t}'(A\underline{X} + \underline{b})\}] \\ &= \exp\{i\underline{t}'\underline{b}\} E[\exp\{i\underline{t}'A\underline{X}\}] \\ &= \exp\{i\underline{t}'\underline{b}\} E[\exp\{i\underline{s}'\underline{X}\}] \quad \text{where } \underline{s} = A^T \underline{t} \\ &= \exp\{i\underline{t}'\underline{b}\} \exp\left\{i\underline{s}'\underline{\mu} - \frac{1}{2}\underline{s}'\Sigma\underline{s}\right\} \\ &= \exp\left\{i\underline{t}'(A\underline{\mu} + \underline{b}) - \frac{1}{2}\underline{t}'(A\Sigma A')\underline{t}\right\}. \end{aligned}$$

By uniqueness property of characteristic function $\underline{Y} \sim \mathcal{N}_q(A\underline{\mu} + \underline{b}, A\Sigma A^\top)$. ■

Example: $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$; $Y = A\underline{X} + \underline{b}$.

$$\underline{Z} = \begin{bmatrix} \underline{Y} \\ \underline{X} \end{bmatrix} \quad \underline{Z} = C\underline{X} + \underline{d} \quad C = \begin{bmatrix} A \\ I_p \end{bmatrix}_{(q+p) \times p} \quad \underline{d} = \begin{bmatrix} \underline{b} \\ 0 \end{bmatrix}$$

$$C\Sigma C^\top = \begin{bmatrix} A\Sigma \\ \Sigma \end{bmatrix} \begin{bmatrix} A^\top & I_p \end{bmatrix} \quad \underline{Z} \sim N_{q+p} \left(\begin{bmatrix} A\underline{\mu} + \underline{b} \\ \underline{\mu} \end{bmatrix}, \begin{bmatrix} A\Sigma A^\top & A\Sigma \\ \Sigma A^\top & \Sigma \end{bmatrix} \right)$$

[3] Expectation and Variance. $E(\underline{X}) = \underline{\mu}$ and $\text{Var}(\underline{X}) = \Sigma$.

Proof: By defn. $\underline{e}_j^\top \underline{X} = X_j \sim \mathcal{N}(\mu_j, \sigma_{jj})$ and

$$(\underline{e}_j + \underline{e}_k)^\top \underline{X} = (X_j + X_k) \sim \mathcal{N}(\mu_j + \mu_k, \sigma_{jj} + \sigma_{kk} + 2\sigma_{jk})$$

From the above two facts, it is clear that—

$$E(X_j) = \mu_j \quad \text{and} \quad \text{Var}(X_j) = \sigma_{jj} \quad \forall j = 1, \dots, p$$

Further,

$$\text{Var}(X_j + X_k) = \text{Var}(X_j) + \text{Var}(X_k) + 2\text{Cov}(X_j, X_k) = \sigma_{jj} + \sigma_{kk} + 2\sigma_{jk}$$

Thus, $\text{Cov}(X_j, X_k) = \sigma_{jk}$. ■

[4] Standardization. Let $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$, and $\Sigma = BB^\top$ for some matrix B . $\Leftrightarrow \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$, where $\underline{Z} \sim \mathcal{N}(\underline{0}, I)$.

Proof: $[\Sigma > 0]$ $\Sigma = U\Lambda U^\top$ with $UU^\top = U^\top U = I_{p \times p}$, and $\Lambda = \text{Diag}(\lambda_1, \dots, \lambda_p)$ $\lambda_j > 0 \forall j$ as $\Sigma > 0$.

Define $\Lambda^{1/2} = \text{Diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p})$

The choices of B can be $B = U\Lambda^{1/2}$ (non-symmetric) or $B = U\Lambda^{1/2}U^\top$ (symmetric).

In both the cases, B^{-1} exists. For symmetric B , $B^{-1} = U\Lambda^{-1/2}U^\top$ where $\Lambda^{-1/2} = \text{Diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_p})$, while for the non-symmetric B , $B^{-1} = \Lambda^{-1/2}U^\top$.

($\Leftarrow$ part) $\underline{Z} \sim N_p(0, I)$

$$B\underline{Z} + \underline{\mu} \sim N_p(\underline{\mu}, BB^\top = \Sigma) \quad \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$$

($\Rightarrow$ part) $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$

$$\underline{Z} = B^{-1}(\underline{X} - \underline{\mu}) \sim N_p(0, I) \implies \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$$

[5] **Independence.** Let $\underline{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \sim \mathcal{N}_p(\underline{\mu}, \Sigma)$, $p_1 + p_2 = p$, then $\underline{X}_1 \perp \underline{X}_2 \Leftrightarrow \Sigma_{12} = \underline{0}$, where

$$\Sigma = \begin{bmatrix} \Sigma_{11}^{p_1 \times p_1} & \Sigma_{12}^{p_1 \times p_2} \\ \Sigma_{21}^{p_2 \times p_1} & \Sigma_{22}^{p_2 \times p_2} \end{bmatrix}, \quad \text{and} \quad \Sigma_{21} = \Sigma'_{12}.$$

Proof: (4 parts) We can easily verify this by property [3] of ch-fn.,

$$\text{as } \phi_{\underline{X}}(\underline{t}) = \exp \left\{ i \underline{t}' \underline{\mu} - \frac{1}{2} \underline{t}' \Sigma \underline{t} \right\} = \exp \left\{ i \underline{t}'_1 \underline{\mu} - \frac{1}{2} \underline{t}'_1 \Sigma_{11} \underline{t}_1 \right\} \times \exp \left\{ i \underline{t}'_2 \underline{\mu} - \frac{1}{2} \underline{t}'_2 \Sigma_{22} \underline{t}_2 \right\}.$$

As $\underline{X}_i \sim \mathcal{N}(\underline{\mu}_i, \Sigma_{ii})$ marginally, $i = 1, 2$, this implies

$$\phi_{\underline{X}}(\underline{t}) = \phi_{\underline{X}_1}(\underline{t}_1) \times \phi_{\underline{X}_2}(\underline{t}_2) \Rightarrow \underline{X}_1 \perp \underline{X}_2.$$

($\Rightarrow$ part) If $\underline{X}_1 \perp \underline{X}_2$, then $\text{Cov}(X_{ij}, X_{2k}) = 0$ for each $j = 1, \dots, p_1$, $k = 1, \dots, p_2$. Thus, $\Sigma_{12} = 0$.

[6] **Density.**

Case I: ($\Sigma > 0$)

By property [4], in this case $\underline{Z} = \Sigma^{-1/2}(\underline{X} - \underline{\mu}) \sim \mathcal{N}(\underline{0}, I_p)$.

By property [5], the p -components of $\underline{Z}$ are independent and identically distributed (IID) following $\mathcal{N}(0, I)$ distribution. Thus,

$$f_{\underline{Z}}(\underline{z}) = (2\pi)^{-p/2} \exp \left\{ -\frac{1}{2} \underline{z}^T \underline{z} \right\}.$$

As the transformation $\underline{Z} \rightarrow \underline{X} = \underline{\mu} + \Sigma^{1/2} \underline{Z}$ is one-one, the density of $\underline{X}$ can be obtained as:

$$\begin{aligned} f_{\underline{X}}(\underline{x}) &= f_{\underline{Z}}(\underline{z}) \Big|_{\underline{Z}=\Sigma^{-1/2}(\underline{X}-\underline{\mu})} \times J_{\underline{Z} \rightarrow \underline{X}} \\ &= |\Sigma|^{-1/2} (2\pi)^{-p/2} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}. \end{aligned}$$

as

$$J_{\underline{Z} \rightarrow \underline{X}} = \left| \det \left(\frac{\partial \underline{Z}}{\partial \underline{X}} \right) \right| = \|B^{-1}\| = \frac{1}{\|B\|} = \frac{1}{|B|}, \quad \text{and} \quad |\Sigma| = |BB^T| = |B|^2 \quad \Rightarrow \quad \frac{1}{|B|} = \frac{1}{|\Sigma|^{1/2}}.$$

Further

$$\sum_{i=1}^p z_i^2 = \underline{z}^T \underline{z} = (\underline{x} - \underline{\mu})^T (B^{-1})^T B^{-1} (\underline{x} - \underline{\mu}) = (\underline{x} - \underline{\mu})^T (BB^T)^{-1} (\underline{x} - \underline{\mu}) = (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}).$$

Equi-probability contour plots.

The contour plots or contours are the collection of points of equal probability density. Thus, the set

of points with density equal to q is:

$$S_q = \left\{ \underline{x} : f_{\underline{x}}(\underline{x}) = q \right\} = \left\{ \underline{x} : (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = c_q \right\},$$

where

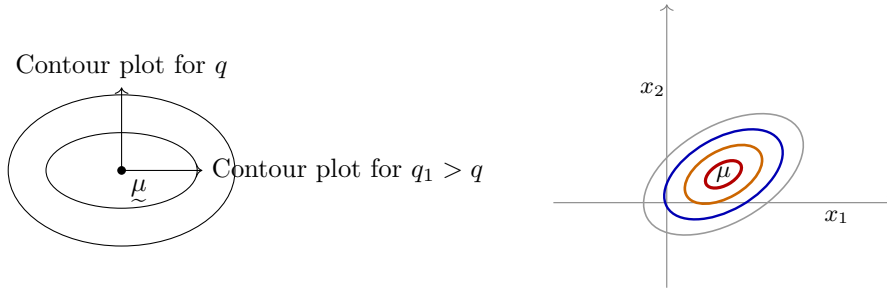
$$c_q = -2 \log \left\{ |\Sigma|^{1/2} (2\pi)^{p/2} q \right\}.$$

Clearly, S_q contains points on an ellipsoid, characterized by parameters $(\underline{\mu}, \Sigma, c_q)$, centered at $\underline{\mu}$. Also, note that the set of high density region $\rho(q) = \left\{ \underline{x} : f_{\underline{x}}(\underline{x}) \geq q \right\}$ is the interior of the ellipse.

For any $q_1 > q$, $S_{q_1} = \left\{ \underline{x} : (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = c_{q_1} \right\}$, where $c_{q_1} < c_q$, and therefore the points form a concentric ellipsoid, with smaller lengths of principal axes.

When $\underline{\mu} = \underline{0}$, and $\Sigma = I$. In this case, $S_q = \left\{ \underline{x} : \underline{x}^T \underline{x} = c_q \right\}$, and thus the equi-probability contours are spherical and centered at $\underline{0}$.

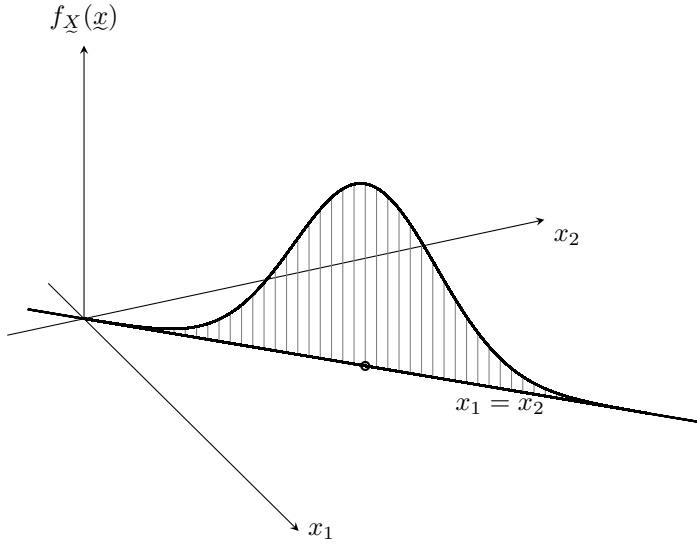
When $\underline{\mu} \neq \underline{0}$, and $\Sigma \neq I$. In this case, S_q is elliptical.



Case II: When Σ is singular, then density exists only in hyperplane which is orthogonal to the null-space of Σ , i.e., in the row-space of Σ .

An example of the case with $\det(\Sigma) = 0$: Let $\underline{X} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right)$. Here, X_1 and X_2 are perfectly correlated. We will show that here $P(X_1 = X_2) = 1$.

Let $\underline{a} = (1, -1)^T$, then $\underline{a}^T \underline{X} = X_1 - X_2 \sim N(0, 0)$, which implies that $E(X_1 - X_2) = 0$ and $var(X_1 - X_2) = 0$, which happen only if $P(X_1 = X_2) = 1$. Then density of $\underline{X}$ exists on the hyperplane $x_1 = x_2$, as $(1, -1)\Sigma = (0 \ 0)$.



In general, if Σ is singular, then the density will exist on the line $\Sigma y = 0$ ($y \neq 0$), as

$$y^T X \sim N(y^T \mu, y^T \Sigma y), \quad \text{which implies} \quad P(y^T X = 0) = 1$$

.

[7] Independence of two linear forms. Let $\underline{X} \sim N(\mu, \Sigma)$; $\Sigma \geq 0$. For what choices of A and B , $\underline{Y} = A\underline{X} \perp B\underline{X} = \underline{Z}$?

First, we see that $(\underline{Y}, \underline{Z})^\top$ is jointly normally distributed, as

$$\begin{bmatrix} \underline{Y} \\ \underline{Z} \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} \underline{X} = C\underline{X} \sim N(C\underline{\mu}, C\Sigma C^\top).$$

Therefore, $\underline{Y} \perp \underline{Z}$, if and only if $\text{Cov}(\underline{Y}, \underline{Z}) = 0$.

When is $\text{Cov}(\underline{Y}, \underline{Z}) = 0$? Answer: $A\Sigma B' = 0$. Hence $A\Sigma B' = 0 \iff AX \perp BX$.

[7] (Marginal and Conditional distributions)

Marginal: From property (2), it is straightforward to obtain the marginal distributions. Let

$$\underline{X} = \begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix},$$

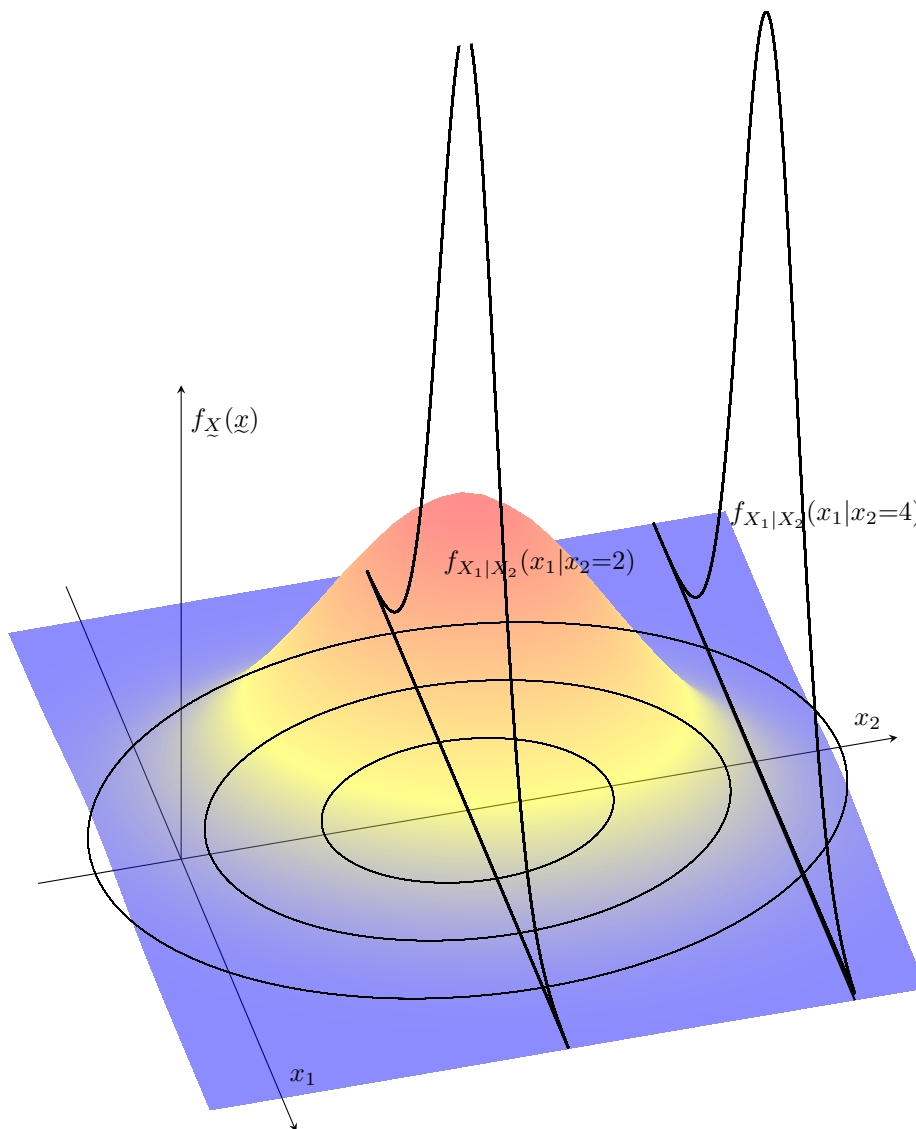
then the marginal of $\underline{X}_2 = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \underline{X}$ is normally distributed with parameters

$$\underline{\mu}_2 \text{ and } \Sigma_{22} = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} 0 \\ I_{p_2} \end{pmatrix}.$$

Conditional distribution: (when $\Sigma_{22} > 0$)

Question. What is the conditional distribution of $\underline{X}_1$ given $\underline{X}_2 = \underline{x}_2$?

It is the distribution of $\underline{X}_1$ informed by the value of $\underline{X}_2 = \underline{x}_2$.



To obtain this, we follow the steps below:

- i. We will find B such that $\underline{Z} = \underline{X}_1 - B\underline{X}_2 \perp \underline{X}_2$.

As $\underline{Z} \perp \underline{X}_2$, the conditional distribution of $\underline{Z}$ given $\underline{X}_2 = \underline{x}_2$ is the same as the marginal distribution.

- ii. Using this fact and the marginal distribution of $\underline{Z}$, we will find the conditional distribution of $\underline{Z} + B\underline{X}_2 (= \underline{X}_1)$ given $\underline{X}_2 = \underline{x}_2$.

Step i. Define $\underline{Z} := [I_{p_1} \ -B] \begin{bmatrix} \underline{X}_1 \\ \underline{X}_2 \end{bmatrix}$, $\underline{X}_2 = [0 \ I_{p_2}] \begin{bmatrix} \underline{X}_1 \\ \underline{X}_2 \end{bmatrix}$

$$[I_{p_1} \ -B] \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0 \iff [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0 \iff \Sigma_{12} - B\Sigma_{22} = 0$$

$$\iff B = \Sigma_{12}\Sigma_{22}^{-1}$$

Therefore, for $\underline{Z} = \underline{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{X}_2$, we have $\underline{Z} \perp \underline{X}_2$, and $\underline{Z} \sim N(\underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{\mu}_2, \Sigma_{11.2})$, where

$$\begin{aligned} \Sigma_{11.2} &= [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} I_{p_1} \\ -B' \end{bmatrix} = \Sigma_{11} - B\Sigma_{21} - \Sigma_{12}B' + B\Sigma_{22}B' \\ &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}. \end{aligned}$$

Step ii. Given $\underline{X}_2 = \underline{x}_2$, $\underline{Z}$ has same distribution as its marginal. Hence, $\underline{X}_2 = \underline{x}_2$,

$$\underline{X}_1 = \underline{Z} + \Sigma_{12}\Sigma_{22}^{-1}\underline{x}_2 \sim N(\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2), \Sigma_{11.2})$$

1.4 Distribution of Quadratic Forms

Definition 1 (Idempotent Matrix). A is idempotent if $A^2 = A$.

A is symmetric idempotent $\iff$ the eigenvalues of A are 1 with multiplicity $\rho(A)$ [rank of A] and 0 with multiplicity $p - \rho(A)$.

Property: $\rho(A) = \text{tr}(A)$.

Result 1 (Cochran's Theorem). Let $\underline{Y} \sim N(0, I_p)$ and let A be a symmetric matrix.

If A is idempotent then

$$\underline{Y}^T A \underline{Y} \sim \chi_{(q)}^2 \quad \text{where } q = \rho(A).$$

Proof Consider the spectral decomposition of A as follows:

$$A = U \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} U^T = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} = U_1 U_1^T \quad \text{where } UU^T = U^T U = I$$

$$\begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \begin{bmatrix} U_1 & U_2 \end{bmatrix} = \begin{bmatrix} U_1^T U_1 & U_1^T U_2 \\ U_2^T U_1 & U_2^T U_2 \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{p-q} \end{bmatrix} \Rightarrow \begin{cases} U_1^T U_1 = I, \\ U_1^T U_2 = 0, \quad U_2^T U_2 = I. \end{cases}$$

Now

$$\underline{Y}^T A \underline{Y} = (U_1^T \underline{Y})^T (U_1^T \underline{Y}) = \|\underline{Z}\|^2 \quad \text{where } \underline{Z} = U_1^T \underline{Y} \sim N_q(0, I)$$

Therefore,

$$= \sum_{i=1}^q Z_i^2 \sim \chi_{(q)}^2 \iff Z_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

Example. Let $\underline{Y} \sim N_n(0, \sigma^2 I)$, $S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$.

To show that $\frac{(n-1)S_y^2}{\sigma^2} \sim \chi_{(n-1)}^2$. Observe that

$$(n-1)S_y^2 = \sum_{i=1}^n Y_i^2 - n\bar{Y}^2, \quad n\bar{Y} = \sum_{i=1}^n Y_i = \underline{Y}^T \underline{1}.$$

Therefore

$$(n-1)S_y^2 = \underline{Y}^T \underline{Y} - \frac{n}{n^2} \underline{Y}^T \underline{1} \underline{1}^T \underline{Y} = \underline{Y}^T \left(I - \frac{1}{n} \underline{1} \underline{1}^T \right) \underline{Y}.$$

Let $H_n = \left(I - \frac{1}{n} \underline{1} \underline{1}^T \right)$, then $H_n^2 = H_n$ (verify), and $\text{tr}(H_n) = \rho(H_n) = q = n-1$. Therefore, by the previous theorem

$$\underline{Z} = \frac{1}{\sigma} \underline{Y} \sim N(\underline{0}, I), \quad \text{and} \quad \frac{(n-1)S_y^2}{\sigma^2} = \underline{Z}^T H_n \underline{Z} = \frac{\underline{Y}^T H_n \underline{Y}}{\sigma^2} \sim \chi_{(n-1)}^2.$$

Result 2. $\underline{Y} \sim N_p(\underline{0}, \Sigma)$, A is symmetric and $A\Sigma$ has eigenvalues 0 or 1.

Then $\underline{Y}^T A \underline{Y} \sim \chi_r^2$ where $r = \rho(A\Sigma)$.

Proof. $\Sigma = BB^T$ and $\underline{Y} \stackrel{D}{=} B\underline{Z}$ where $\underline{Z} \sim N(\underline{0}, I)$.

$$\underline{Y}^T A \underline{Y} \stackrel{D}{=} \underline{Z}^T B^T A B \underline{Z}.$$

Note that the non-zero eigenvalues of $B^T A B$ are the same as that of $AB B^T = A\Sigma$.

So, $B^T A B$ has the eigenvalue 1 with multiplicity $r = \rho(A\Sigma)$, and $(p-r)$ zero-eigenvalues.

This implies that the spectral decomposition of $B^T A B$ will be of the form:

$$B^T A B = U U^T \quad \text{with} \quad U^T U = I_r.$$

Thus,

$$\underline{Y}^T A \underline{Y} = \underline{Z}^T B^T A B \underline{Z} = \underline{Z}^T U U^T \underline{Z} = \|U^T \underline{Z}\|^2.$$

$$\underline{Z} \sim N(\underline{0}, I) \Rightarrow U^T \underline{Z} \sim N(\underline{0}, I_r) \Rightarrow \|U^T \underline{Z}\|^2 \sim \chi_r^2.$$

□

Result 3 (Homework). $Y \sim N_p(\mu, \Sigma)$, $\Sigma > 0$. Then $(Y - \mu)^T \Sigma^{-1} (Y - \mu) \sim \chi_p^2$.

Result 4 (Craig's Theorem). $Y \sim N_p(0, I)$ and A, B are two symmetric and idempotent matrices.

Then $Y^T A Y \perp\!\!\!\perp Y^T B Y$ iff $AB = 0$.

Proof. $Y^T A Y = \|AY\|^2$ and $Y^T B Y = \|BY\|^2$.

Enough to show that $AY \perp\!\!\!\perp BY \Leftrightarrow AB = 0$. [apply Property 6 of normal distribution]

□

Example (continued). Recall that $X \sim N_n(\mu 1, \sigma^2 I)$, $\sigma^2 > 0$, and

$$S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (n-1)S_y^2 = Y^\top H_n Y. \quad \text{where } H_n = \left(I - \frac{1}{n} 11^\top \right) \text{ is idempotent.}$$

Let $Z = \sigma^{-1}(Y - \mu 1) \sim N(0, I_n)$, then $Z^\top H_n Z \sim \chi_{n-1}^2$, that is, $\sigma^{-2}(Y - \mu 1)^\top H_n (Y - \mu 1) \sim \chi_{n-1}^2$.

Now,

$$(Y - \mu 1)^\top H_n (Y - \mu 1) = Y^\top H_n Y - 2\mu 1^\top H_n Y + \mu^2 1^\top H_n 1 = Y^\top H_n Y,$$

as $H_n 1 = 0$ then we get the same conclusion

$$\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Chapter 2

Linear Models (Revisit)

Linear Regression

Recall the linear regression model with response y , and regressors/covariates $\{x_1, \dots, x_p\}$ and n observations as follows:

$$\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where

$$\underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

We assume $\varepsilon_1, \dots, \varepsilon_n$ are independent and identically distributed (i.i.d.), $E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}$ and $\text{var}(\underset{\sim}{\varepsilon}) = \sigma^2 I$.

Parameters in the linear regression model: $(\underset{\sim}{\beta}, \sigma^2)$, where

- $\underset{\sim}{\beta}$: vector of regression coefficients. In particular, β_j measures the marginal increment in y when the covariate x_j is increased by one unit, keeping other covariates fixed at a certain level, $j = 1, \dots, p$. Further, the intercept β_0 is the mean effect of y , when all the covariates are set to 0.
- σ^2 : error variance. This parameter measures the average deviation of y from the explainable part (i.e., the part of y which can be systematically explained by $\{x_1, \dots, x_p\}$).

Difference between the interpretations of the multiple regression model, compared to individual simple regression models. Consider the follow to system of models:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i, \quad i = 1, \dots, n, \quad (2.1)$$

and

$$\begin{cases} Y_i = \gamma_0 + \gamma_1 x_{1i} + \varepsilon_{1i} \\ \vdots \\ Y_i = \gamma_0 + \gamma_p x_{pi} + \varepsilon_{pi} \end{cases} \quad (2.2)$$

In (2.1), the parameters $\{\beta_1, \dots, \beta_p\}$ measures the effect of $\{x_1, \dots, x_p\}$ on y . In (2.2), the parameters $\{\gamma_1, \dots, \gamma_p\}$ measures the effect of $\{x_1, \dots, x_p\}$ on y .

Q. Does the two set of estimates have same interpretation? Also, will the least squares estimates of β_j be same as that of γ_j ?

A. No. Here β_j signifies the effect of x_j keeping other covariates fixed at a certain level, whereas, γ_j signifies the effect of x_j irrespective of the other covariates. For example, suppose $p = 2$, for some commodity $x_1 \equiv \text{demand}$, $x_2 \equiv \text{supply}$ and $y \equiv \text{price}$. Here $\gamma_1 = 2$ implies that if demand increases by one unit then price increases by 2 unit, irrespective of the supply. However, $\beta_1 = 4$ indicates, if demand increases by one unit, but the supply remains fixed at a certain level, then price increase by 4 unit. Observe that, in this example, it is in fact intuitive that $\beta_1 > \gamma_1$.

Simpson's Paradox: This is a statistical phenomenon where the sign of β_j and γ_j reverses. This occurs because the second variable is related to both the first variable and the response, masking the true relationship. For example, suppose

$$y = \begin{cases} 1 & \text{an exam is taken} \\ & \text{in the class,} \\ 0 & \text{otherwise,} \end{cases}, \quad x_1 = \text{class attendance}, \quad x_2 = \begin{cases} 1 & \text{the class activity is included} \\ & \text{in final grading,} \\ 0 & \text{otherwise.} \end{cases},$$

and we have data on all classes in all days during the semester. Usually exams are graded, so marginally, γ_1 will be positive indicating that more students appears in a date of exam. However, if we keep x_2 fixed, then β_1 may be negative, since students tend to avoid exams, if possible the topic is not relevant for final grading.

2.1 Least Squares Method

Here we find an estimate of β by minimizing

$$\varepsilon_{\sim}^T \varepsilon_{\sim} = (Y_{\sim} - X_{\sim} \beta)^T (Y_{\sim} - X_{\sim} \beta) = \sum_{i=1}^n (Y_i - \beta_0 - \cdots - \beta_p x_{pi})^2 = S(\beta),$$

w.r.t. $\beta_{\sim}$.

To minimize $S(\beta) : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$ we differentiate $S(\beta)$ w.r.t. $\beta_{\sim}$.

$$\frac{\partial S(\beta)}{\partial \beta_{\sim}} = \frac{\partial}{\partial \beta_{\sim}} \left(\|Y_{\sim}\|^2 + \beta_{\sim}^T X^T X \beta_{\sim} - 2Y_{\sim}^T X \beta_{\sim} \right) = 0 + 2X^T X \beta_{\sim} - 2X^T Y_{\sim}.$$

First order condition:

$$\frac{\partial S(\beta)}{\partial \beta_{\sim}} = 0 \Rightarrow X^T X \beta_{\sim} = X^T Y_{\sim}.$$

$$\hat{\beta}_{\sim} = (X^T X)^{-1} X^T Y_{\sim} \quad \text{when } (X^T X)^{-1} \text{ exists.}$$

Second order condition:

$$\frac{\partial^2 S(\beta)}{\partial \beta_{\sim} \partial \beta_{\sim}^T} = X^T X.$$

When $(X^T X)^{-1}$ exists, then $X^T X$ is positive definite, $\Rightarrow \hat{\beta}_{\sim}$ is the unique minima.

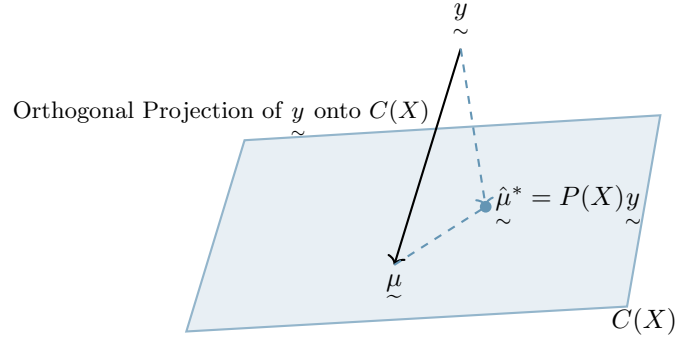
Useful derivatives: Derive the following:

1. $\frac{\partial}{\partial \underline{x}} (\underline{x}^T A \underline{x}) = \frac{\partial}{\partial x_j} \left(\sum_{i,k} a_{ik} x_i x_k \right)_{j=1,\dots,p} = (A + A^T)x = 2Ax \quad (\text{when } A \text{ is symmetric}).$
2. $\frac{\partial a^T \underline{x}}{\partial \underline{x}} = \underline{x}.$
3. $\frac{\partial A \underline{x}}{\partial \underline{x}} = A$ when A is symmetric.

Geometric interpretation

The problem is to find $\hat{\beta}_{\sim}$ s.t. $\|y - X\hat{\beta}_{\sim}\|^2$ is minimum.

i.e., to find the component $\hat{\mu}^*$ in the column space of X , $C(X)$, which is closest to y , where $C(X) = \{\underline{z} : \underline{z} = a_0 \mathbf{1}_{\sim(0)} + a_1 \underline{x}_{\sim(1)} + \dots + a_p \underline{x}_{\sim(p)} = X \underline{a}_{\sim}\}.$



Orthogonal projection of y onto $C(X)$; $C(X)$ is the plane.

We will show that $\hat{\mu}_{\sim}^* = \hat{\mu}_{\sim\text{OLS}}$ is the closest point of y in $C(X)$.

Properties of Orthogonal projection matrix

Let $P(X)$ be the orthogonal projection matrix onto the column space of X . Then the following properties are satisfied:

- $P^2(X) = P(X) \Leftrightarrow P(X)[I - P(X)] = 0$ and also symmetric.
- If $\mu \in C(X)$ then $P(X)\mu = \mu$.
- $\rho(P(X)) = \rho(X) = \text{tr}(P(X))$.
- If $(X'X)^{-1}$ exists then $P(X) = X(X'X)^{-1}X'$.
If $(X'X)$ is singular then $P(X) = X(X'X)^-X'$, where A^- is any g-inverse of A .

Least Square solution (Geometrically)

Let

$$\hat{\mu}_{\sim}^* = \hat{\mu}_{\sim\text{OLS}} = P(X)y \in C(X).$$

We will show that $\hat{\mu}_{\sim}^*$ is the closest point of y in $C(X)$.

Take any $\mu \in C(X)$. Observe that

$$\|y - \mu\|^2 = \|y - \hat{\mu}_{\sim}^* + \hat{\mu}_{\sim}^* - \mu\|^2 = \|y - \hat{\mu}_{\sim}^*\|^2 + \|\hat{\mu}_{\sim}^* - \mu\|^2.$$

The product term vanishes due to the following:

$$\begin{aligned}
 (\underset{\sim}{y} - \underset{\sim}{\hat{\mu}^*})^T (\underset{\sim}{\hat{\mu}^*} - \underset{\sim}{\mu}) &= (\underset{\sim}{y} - P(X)\underset{\sim}{y})^T (P(X)\underset{\sim}{y} - \underset{\sim}{\mu}) \\
 &= \underset{\sim}{y}^T (I - P(X)) (P(X)\underset{\sim}{y} - \underset{\sim}{\mu}) \\
 &= \underset{\sim}{y}^T \underbrace{(I - P(X))P(X)}_0 \underset{\sim}{y} - \underset{\sim}{y}^T \underbrace{(I - P(X))}_{\underset{\sim}{\mu} - P(X)\underset{\sim}{\mu}} \underset{\sim}{\mu} \\
 &= 0 \quad \text{on } \underset{\sim}{\mu} \in C(X).
 \end{aligned}$$

Thus, the distance is minimized if $\underset{\sim}{\hat{\mu}^*} = \underset{\sim}{\hat{\mu}}_{\text{OLS}} = \underset{\sim}{\mu}$. So, the optimum point in $C(X)$ is $\underset{\sim}{\hat{\mu}^*}$.

Q. For which choice of $\underset{\sim}{\beta}$, $P(X)\underset{\sim}{y} = X\underset{\sim}{\beta}$? That is, what is the solution to the following equation

$$\Rightarrow X(X'X)^-X'\underset{\sim}{y} = X\underset{\sim}{\beta} \quad (\star)$$

Case 1: $(X'X)^{-1}$ exists

$(X'X)^{-1}$ exists $\Leftrightarrow X$ has full column rank, $\rho(X) = p + 1$

$\Leftrightarrow$ There exists a matrix Z such that $ZX = I_{p+1}$ (i.e., left inverse of X exists)

So, pre-multiply $(\star)$ by Z ,

$$\underbrace{ZX}_{I_{p+1}}(X'X)^{-1}X'\underset{\sim}{y} = \underbrace{ZX}_{I_{p+1}}\underset{\sim}{\beta} = \hat{\underset{\sim}{\beta}}_{\text{OLS}}.$$

Case II: $X'X$ is singular

Recall the problem of finding solution of the system of equations $A\underset{\sim}{z} = \underset{\sim}{b}$ where $\rho(A) < q$, q being the number of columns of A . In this case, a solution exists if $\underset{\sim}{b} \in C(A)$, and as $\rho(A) < q$, there are infinite solutions.

Here $A = X$ and $\underset{\sim}{z} = \underset{\sim}{\beta}$ and $\underset{\sim}{b} = P(X)\underset{\sim}{y} \in C(X)$. So, a solution exists. Further, as $\rho(X) < p + 1$, the solution is not unique.

One possible solution is

$$\hat{\underset{\sim}{\beta}} = (X'X)^-X'\underset{\sim}{y}.$$

Definition 2 (Residual). The minimized error after inserting the least square solution is called the **residual**, and is denoted by $\underset{\sim}{e} = \underset{\sim}{y} - X\hat{\underset{\sim}{\beta}}_{\text{OLS}} = [I - P(X)]\underset{\sim}{y}$.

The minimized error sum of squares, equivalently, the minimized $\underset{\sim}{\varepsilon}^T \underset{\sim}{\varepsilon}$ is called the **residual sum of squares** (RSS)

$$\text{RSS} = \|\underset{\sim}{y} - X\hat{\underset{\sim}{\beta}}_{\text{OLS}}\|^2 = \underset{\sim}{y}^T (I - P(X))\underset{\sim}{y} \quad (\text{RSS residual Sum of Squares}).$$

Note that, $\underset{\sim}{e}$ and $\underset{\sim}{\varepsilon}$ are NOT the same. While $\underset{\sim}{\varepsilon}$ is an unobservable quantity indicating the part of $\underset{\sim}{y}$ which can not be explained by a linear function of $\{x_1, \dots, x_p\}$, given the data, $\underset{\sim}{e}$ is an observable

quantity indicating the (least-square) minimized estimation of ε .

Remark: When $X'X$ is singular, then the solution to least squares equation (i.e., β in $(*)$) is not unique.

However, $P(X)$ is uniquely defined in this case as well. Therefore, the terms depending on $P(X)$, such as $\hat{\mu}_{\text{OLS}} = P(X)\underline{y}$, $\underline{\varepsilon} = (I - P(X))\underline{y}$ and RSS are also unique even if $X'X$ is singular.

Estimation of σ^2

Result

$\frac{\text{RSS}}{n - r}$, where $\rho(X) = r$, is an unbiased estimator of σ^2 .

Proof. Observe that

$$\begin{aligned} E\left(\frac{\text{RSS}}{n - r}\right) &= E\left(\frac{\underline{y}^T (I - P(X)) \underline{y}}{n - r}\right) \\ &= \frac{1}{n - r} \left\{ (X\beta)^T (I - P(X)) (X\beta) + \text{tr}(\sigma^2 I (I - P(X))) \right\} \end{aligned}$$

[As $X\beta \in e(X)$, $(I - P(X))X\beta = 0$ indicating that the first term in the above equation is 0.]
Also, recall from PS 1 that $E(\underline{x}^T A \underline{x}) = \underline{\mu}^T A \underline{\mu} + \text{tr}(A\Sigma)$ when $E(\underline{x}) = \underline{\mu}$, $\text{var}(\underline{x}) = \Sigma$. Thus,

$$E\left(\frac{\text{RSS}}{n - r}\right) = \frac{1}{n - r} \left\{ 0 + \sigma^2 (\text{tr}(I_n) - \text{tr}(P(X))) \right\} = \sigma^2,$$

since $\text{tr}(P(X)) = \rho(P(X)) = \rho(X) = r$. □

We write $\hat{\sigma}^2 = \frac{\text{RSS}}{n - r}$.

Properties of $\hat{\beta}_{\text{OLS}}$

Result 5. Let X have full column rank, and $\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y$. Then

- i) $E_{\beta}(\hat{\beta}_{\text{OLS}}) = \beta$.
- ii) $\text{Var}_{\beta}(\hat{\beta}_{\text{OLS}}) = \sigma^2 (X'X)^{-1}$.
- iii) Let $\underline{c}^T \underline{\mu} = g(\underline{\mu})$ is the parameter of interest, where $\underline{\mu} = X\beta$, and $\hat{\underline{\mu}}_{\text{OLS}} = X\hat{\beta}_{\text{OLS}}$. Then $\underline{c}^T \hat{\underline{\mu}}_{\text{OLS}}$ is the (unique) BLUE (best linear unbiased estimator) of $g(\underline{\mu})$
[This result remains unchanged when $X'X$ is singular.]

Proof. i) Trivial.

ii) Observe that

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \text{Var}((X'X)^{-1}X'y) = (X'X)^{-1}X' \text{var}(y) X(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

iii) Observe that, $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ is a linear unbiased estimator, as

$$\mathcal{C}^T \hat{\mu}_{\text{OLS}} = \mathcal{C}^T X(X'X)^{-1}X'y \quad (\text{linear in } y),$$

and

$$E(\mathcal{C}^T \hat{\mu}_{\text{OLS}}) = \mathcal{C}^T \underbrace{X(X'X)^{-1}X'}_{P(X)} \underbrace{X\beta}_{\mu \in C(X)} = \mathcal{C}^T \mu \quad (\text{unbiased}).$$

It remains to show that $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ has the lowest variance among all other linear unbiased estimators.

Let $\mathcal{Q}^T y$ be another LUE of $\mathcal{C}^T \mu$. As $\mathcal{Q}^T y$ is unbiased, $E(\mathcal{Q}^T y) = \mathcal{Q}^T \mu = \mathcal{C}^T \mu$.

$$\begin{aligned} \Rightarrow (\mathcal{Q} - \mathcal{C})^T \mu &= (\mathcal{Q} - \mathcal{C})^T X\beta = 0 \\ \Rightarrow (\mathcal{Q} - \mathcal{C})^T X &= \mathbf{0}^T \quad (\text{as } \beta \in \mathbb{R}^{p+1} \text{ is any arbitrary vector}) \\ \Rightarrow (\mathcal{Q} - \mathcal{C})^T P(X) &= \mathbf{0}^T \\ \Rightarrow P(X)(\mathcal{Q} - \mathcal{C}) &= \mathbf{0} \quad (*). \end{aligned}$$

Now, consider the difference in variance of $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ compared to $\mathcal{Q}^T y$ as

$$\begin{aligned} \text{Var}(\mathcal{Q}^T y) - \text{Var}(\mathcal{C}^T \hat{\mu}_{\text{OLS}}) &= (\mathcal{Q}^T \mathcal{Q})\sigma^2 - \mathcal{C}^T \text{Var}(P(X)y)\mathcal{C} \\ &= \sigma^2 [\mathcal{Q}^T \mathcal{Q} - \mathcal{C}^T P(X)\mathcal{C}] \quad ((P(X)\mathcal{C})^T (P(X)\mathcal{C}) = \mathcal{C}^T P(X)\mathcal{C}) \\ &= \sigma^2 [\mathcal{Q}^T \mathcal{Q} - \mathcal{Q}^T P(X)\mathcal{Q}] \quad (P(X)\mathcal{C} = (P(X)\mathcal{Q}) \text{ by } (*)) \\ &= \sigma^2 \mathcal{Q}^T (I - P(X))\mathcal{Q} = \sigma^2 \|(I - P(X))\mathcal{Q}\|^2 \\ &\geq 0. \end{aligned}$$

Equality holds, $\mathcal{Q} = P(X)\mathcal{Q} \stackrel{(*)}{=} P(X)\mathcal{C}$. Then $\mathcal{Q}^T y = \mathcal{C}^T P(X)y = \mathcal{C}^T \hat{\mu}_{\text{OLS}}$. □

2.2 Distribution Theory

Let us now assume

$$\varepsilon \sim N_n(\mathbf{0}, \sigma^2 I) \Leftrightarrow \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \Leftrightarrow Y = X\beta + \varepsilon \sim N_n(X\beta, \sigma^2 I).$$

Result

Let $\underline{Y} \sim N(X\underline{\beta}, \sigma^2 I)$, and $\rho(X) = p + 1$. Then

- i. $\hat{\underline{\beta}}_{\text{OLS}} \sim N_{p+1}(\underline{\beta}, \sigma^2(X'X)^{-1}) \quad ((X'X)^{-1} \text{ exists}).$
- ii. $\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2.$

Proof. i. Trivial.

ii. $\text{RSS} = \sigma^{-2} \underline{\varepsilon}^T \underline{\varepsilon} = \sigma^{-2} \underline{y}^T (I - P(X)) \underline{y}$. Note that $\underline{Z} = \sigma^{-1} \underline{Y} - X\underline{\beta} \sim N(\underline{0}, I)$.

Therefore,

$$\begin{aligned} \text{RSS} &= \frac{1}{\sigma^2} (\underline{Y} - X\underline{\beta} + X\underline{\beta})^T (I - P(X)) (\underline{Y} - X\underline{\beta} + X\underline{\beta}) \\ &= \frac{\underline{\varepsilon}^T (I - P(X)) \underline{\varepsilon}}{\sigma^2} + 0 \quad [(I - P(X))X\underline{\beta} = 0, \text{ as } X\underline{\beta} = \mu \in C(X)] \\ &= \underline{Z}^T (I - P(X)) \underline{Z}. \end{aligned}$$

Recall that $\underline{Z} \sim N(\underline{0}, I)$, and A is symmetric, idempotent with $\rho(A) = r, \implies \underline{Z}^T A \underline{Z} \sim \chi_r^2$. As $(I - P(X))$ is symmetric, idempotent with $\rho(I - P(X)) = n - \text{tr}(P(X)) = n - (p + 1)$, the result follows. □

HW: What happens if $\rho(X) = r < p + 1$? Show that, in this case $\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-r}^2$.

Maximum Likelihood Estimation

Log-likelihood,

$$\ell(\underline{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (\underline{Y} - X\underline{\beta})^T (\underline{Y} - X\underline{\beta}).$$

Maximizing the likelihood w.r.t. $\underline{\beta}$, is same as minimizing the negative of last part w.r.t. $\underline{\beta}$, which is $\|\underline{Y} - X\underline{\beta}\|^2, \sigma^2 > 0$.

Thus,

$$\ell(\underline{\beta}, \sigma^2) \leq \ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2).$$

Now, we can simply minimize $\ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2)$ w.r.t. σ^2 .

$$\ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2) = h(\sigma^2) = C - \frac{n}{2} \log \sigma^2 - \frac{\text{RSS}}{2\sigma^2}.$$

The first order condition provides

$$\frac{\partial}{\partial \sigma^2} \ell(\hat{\beta}_{\text{OLS}}, \sigma^2) = 0 \quad \Rightarrow \quad \sigma^2 = \frac{\text{RSS}}{n}.$$

Second order conditions provides

$$\begin{aligned} \left. \frac{\partial^2}{\partial (\sigma^2)^2} \ell(\hat{\beta}_{\text{OLS}}, \sigma^2) \right|_{\sigma^2 = n^{-1} \text{RSS}} &= \left. \frac{\partial}{\partial \sigma^2} \left(-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \text{RSS} \right) \right|_{\sigma^2 = n^{-1} \text{RSS}} = \left. \left(\frac{n}{2\sigma^4} - \frac{\text{RSS}}{\sigma^6} \right) \right|_{\sigma^2 = n^{-1} \text{RSS}} \\ &= \left(\frac{n^3}{2 \text{RSS}^2} - \frac{n^3}{\text{RSS}^2} \right) = -\frac{n^3}{2 \text{RSS}^2} < 0. \end{aligned}$$

Therefore, the maximum likelihood estimates of (β, σ^2) are:

$$\hat{\beta}_{\text{ML}} = \hat{\beta}_{\text{OLS}}, \quad \text{and} \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}.$$

While the MLE of β is unbiased, MLE of σ^2 is a biased estimate.